Paper 75 — The Road Less Scheduled

Aaron Defazio, Xingyu (Alice) Yang, Harsh Mehta, Konstantin Mishchenko, Ahmed Khaled, Ashok Cutkosky
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Schedule-free optimization through interpolation and iterate averaging

Ref 105 | S / D / E; R / B | Must read | Original publication

The paper in one minute

Question: Can an optimizer perform competitively without choosing a learning-rate schedule around a known stopping time T?
Model/Method: Couple a fast base sequence, a running-average sequence, and a gradient-evaluation sequence through a momentum-like interpolation.
Finding: Schedule-Free SGD obtains optimal convex stochastic rates under its assumptions, and Schedule-Free AdamW is competitive in broad deep-learning experiments without a prespecified decay schedule.
Why we care: Schedules are not the only way to obtain stable, good final iterates: where gradients are evaluated matters as much as how iterates are averaged.

1. Why schedules are awkward

Cosine decay and related schedules usually need a planned training horizon T. If training stops early or continues longer, the intended step-size trajectory may be inappropriate. Plain constant-step SGD can be good for early progress, while averaging its iterates reduces variance but may lag behind the changing solution. The paper unifies these ideas with *three* sequences, which must not be mixed up:

$$\begin{aligned} y_t &= (1-\beta)z_t + \beta x_t && \text{(where the gradient is measured),} \\ z_{t+1} &= z_t - \gamma \nabla f(y_t;\zeta_t) && \text{(base update),} \\ x_{t+1} &= (1-c_{t+1})x_t + c_{t+1} z_{t+1} && \text{(reported average).} \end{aligned}$$

With $c_{t+1} = 1/(t+1)$, x is a running average of z. Parameter β interpolates between gradients at the base point ($\beta = 0$) and the averaged point ($\beta = 1$).

2. Worked example: the three sequences really differ

Take $f(w) = \frac{1}{2}(w-2)^2$, start at $x_1 = z_1 = 0$, choose $\gamma = 0.5$ and $\beta = 0.8$. At step 1:

$$y_1 = 0, \quad g_1 = y_1 - 2 = -2, \quad z_2 = 0 - 0.5(-2) = 1, \quad x_2 = \tfrac{1}{2}(0) + \tfrac{1}{2}(1) = 0.5.$$

At step 2, the new gradient is evaluated not at $z_2 = 1$ or $x_2 = 0.5$ but at

$$y_2 = 0.2(1) + 0.8(0.5) = 0.6.$$

Therefore $g_2 = -1.4$, $z_3 = 1 + 0.7 = 1.7$, and $x_3 = \frac{2}{3}(0.5) + \frac{1}{3}(1.7) = 0.9$. The base moves fast; the average lags and stabilizes; the gradient point interpolates between them.

How to think about the algorithm
The algorithm is not merely "SGD with a constant learning rate." Its defining feature is *where* the gradient is evaluated and *how* the resulting base steps are averaged.

3. Theorem: what is guaranteed and what is not

Theorem 1 studies an i.i.d. stochastic convex problem with G-Lipschitz losses, a minimizer $x^\star$, and $D = \|x_1 - x^\star\|$. For the theorem's horizon-dependent choice $\gamma = D/(G\sqrt{T})$, any $\beta \in [0, 1]$ yields

$$\boxed{\mathbb{E}\big[F(x_T) - F(x^\star)\big] \le \frac{DG}{\sqrt{T}}.}$$

This is a stochastic *convex* guarantee for the averaged sequence x_T; it is not a proof that arbitrary deep neural networks converge to a global minimum. It may seem ironic that the theorem chooses γ using T. The answer is that this gives a clean worst-case proof, while the practical schedule-free motivation and algorithms explore using a fixed rate without specifying a decay horizon.

4. Why averaging and the gradient location must be coupled

Polyak–Ruppert averaging uses an average x_t of base iterates z_t but computes gradients at z_t. In the paper's notation, this is $\beta = 0$. Primal averaging computes gradients at x_t, corresponding to $\beta = 1$. Schedule-Free uses intermediate β values to gain some averaging stability while remaining responsive to new gradients. Theorem 2 unifies these choices through an online-to-batch conversion bound: a weighted average's optimization error can be controlled by an online-gradient regret term. This is the theoretical bridge between scheduling, averaging, and momentum.

What happens in real optimizers?

Schedule-Free AdamW adapts the mechanism to an Adam-like base optimizer and evaluates/test-uses the proper averaged sequence. The paper also discusses how momentum allows aggressive step sizes on quadratic models. These implementation details are separate from Theorem 1's abstract convex setting.

Practical caveat
During training, gradients may be evaluated at y_t, but evaluation typically uses the averaged weights x_t. Accidentally reporting the wrong sequence changes what algorithm is being assessed.

5. Findings and main contributions

Contributions. (i) A three-sequence schedule-free update; (ii) an online-to-batch theory unifying interpolation and averaging; (iii) convex convergence guarantees independent of the particular fixed interpolation parameter β; and (iv) broad empirical comparisons showing strong training results for Schedule-Free AdamW. **Evidence.** Figures 2–4 examine optimization behavior across tasks, including language modeling and the AlgoPerf suite. The authors report competitiveness with tuned schedule-based baselines and describe Schedule-Free AdamW as a major part of their winning 2024 AlgoPerf self-tuning submission. These are empirical comparisons, not universal guarantees.

6. Assumptions and limitations

- The flagship $DG/\sqrt{T}$ theorem requires convex Lipschitz losses and i.i.d. sampling; non-convex deep-learning outcomes are empirical.
- A fixed learning rate still must be selected and may need warmup or tuning. "Schedule free" means no prescribed decay schedule, *not* no learning-rate hyperparameters.
- Theory's choice $\gamma = D/(G\sqrt{T})$ uses a training horizon; practical schedule-free behavior is a separate design claim.
- The work is not principally a strict-saddle-escape method, nor does it prove global optimization or generalization for all networks.

7. Quick reference

Base weights	z_t: updated using the sampled gradient.
Average weights	x_t: the running, reported/evaluation sequence.
Gradient point	$y_t = (1-\beta)z_t + \beta x_t$.
Step	$z_{t+1} = z_t - \gamma \nabla f(y_t; \zeta_t)$.
Convex guarantee	$\mathbb{E}[F(x_T) - F(x^\star)] \leq DG/\sqrt{T}$ (Theorem 1).
Not proved	General non-convex global convergence or saddle escape.

Survey bridge / Papers Master note

Schedule-Free SGD/AdamW replaces a predefined learning-rate decay with coupled base, average and gradient-evaluation sequences. Theorem 1 provides an optimal convex stochastic rate, and broader neural-network benefits are empirical. Useful when contrasting learning-rate schedules with dynamic update modifications and Lookahead.